

Letting Mass Come and Go

NubOT: neural unbalanced optimal transport for forecasting single-cell drug responses when cells are born and die

Optimal transport (OT) has quietly become one of the most useful tools in machine learning for comparing two distributions. Given a source population and a target population, it finds the most economical way to move the first onto the second, and in doing so it hands back something valuable: a coupling that says which source points correspond to which target points. That correspondence is exactly what you want when you cannot track individuals directly and can only observe two unpaired snapshots of a population.

But classical OT carries an assumption that is easy to overlook: mass is conserved. Every unit that leaves the source must land somewhere in the target, and nothing is created or destroyed along the way. That is fine when the two populations are simply reshaped versions of each other. It fails precisely when the size of the population changes, which is the situation that matters most in some of the fields where OT is now being applied.

Single-cell biology is the clearest example. Profiling a cell often destroys it, so the same cell can never be measured twice; correspondences across time have to be inferred from unpaired replicas of the population. And under a drug, cells do not just move in feature space, they proliferate and die, so the population grows and shrinks. This NeurIPS 2022 paper introduces NubOT, a neural method for unbalanced optimal transport built on the formalism of semi-couplings. Through a cycle-consistent training scheme, NubOT learns both where cells move and how their mass should rescale, letting it explicitly account for the creation and destruction of mass.

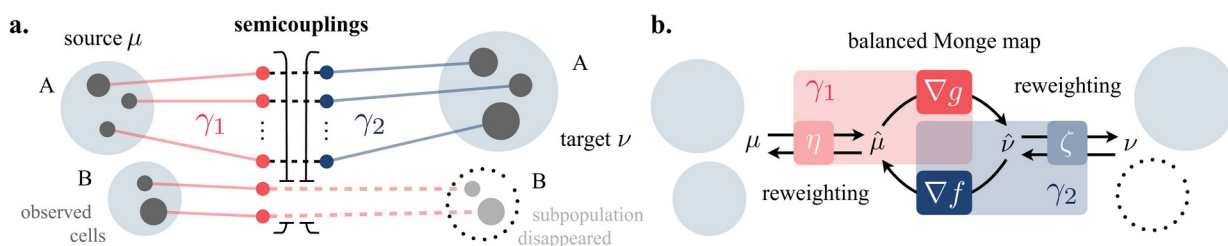


Figure 1. (a) A semi-coupling pair (γ_1, γ_2) together solves the unbalanced OT problem: γ_1 describes where mass goes as it leaves the source, γ_2 where it comes from as it arrives in the target, so a vanishing subpopulation can simply be left unmatched. (b) NubOT parameterizes the semi-couplings as reweighting functions (η, ζ) composed with balanced Monge maps given by the gradients of dual potentials f and g .

Paper: <https://arxiv.org/abs/2209.15621> Code: <https://github.com/FrederikeLuebeck/nubot>

The problem: transport when mass is not conserved

A recurring problem across the natural sciences is modeling how a population changes after an intervention, when you can only observe unpaired snapshots rather than follow individuals. In single-cell biology this is unavoidable. Optimal transport offers a principled way to recover the

missing correspondences by learning an optimal coupling between the source and target distributions, and neural OT methods have made this scalable by learning the transport map itself as a neural network.

The catch is that cellular responses to drugs are heterogeneous. Different cell types and states can respond in opposite directions, with some subpopulations proliferating while others die, so the total number of cells shifts between measurements. Capturing this needs nonlinear maps at the level of single cells, not aggregate averages. Prior neural OT methods either assume the balanced setting where mass is conserved, or, like the state-of-the-art unbalanced GAN approach, only capture the broad trend of growth and shrinkage without recovering the exact reweighting each subpopulation needs. What has been missing is a method that jointly and accurately models where mass moves and how much of it is created or destroyed.

NubOT: splitting transport from reweighting

NubOT builds on the theory of semi-couplings, which relaxes the single coupling of balanced OT into a pair. As shown in Figure 1a, one coupling γ_1 describes where mass goes as it leaves the source, and a second coupling γ_2 describes where it comes from as it arrives in the target. Because the two are allowed to disagree, a subpopulation can shed mass, or gain it, without being forced to send every unit somewhere. The contribution of the paper is to make this formalism learnable and scalable, and to pair it with a training scheme that generalizes to previously unseen cells at test time.

The key idea is to split the hard unbalanced problem into two coupled pieces: feature transformation and mass rescaling. NubOT introduces proxy measures that are simply reweighted versions of the source and target, weighted by scalar fields η and ζ and chosen so that the two proxies have equal mass. Between these now-balanced proxies it solves a standard OT problem, learning forward and backward Monge maps as the gradients of convex potentials represented by an input convex neural network (ICNN). From the learned reweighting functions and maps, the original semi-couplings can then be recovered exactly.

Because the marginals themselves shift every time the reweighting functions update, NubOT cannot solve the two pieces in one shot. Instead it alternates: it estimates the mass rescaling with an efficient single-step unbalanced Sinkhorn update, then updates the transport potentials through a min-max objective, and cycles between the two. This is where cycle-consistency enters, tying the forward and backward maps together so the alternation converges to a coherent solution. Crucially, the reweighting functions are learned as neural networks, so at test time the model can take a new cell, map it to its predicted future state, and simultaneously predict whether its mass should grow or shrink.

A synthetic stress test

Before touching real data, the authors build a controlled benchmark: a two-dimensional mixture of three Gaussian clusters, where the target keeps the same clusters but changes their relative proportions and shifts them in space, across three scenarios of increasing imbalance. The task is to map source to target while getting both the locations and the per-cluster mass right. Figure 2 compares NubOT against the unbalanced GAN baseline. Both methods place the

transported samples in roughly the correct locations, but only NubOT recovers the reweighting each cluster needs, creating mass where the target cluster is denser and destroying it where it is sparser, without leaking mass between non-corresponding clusters.

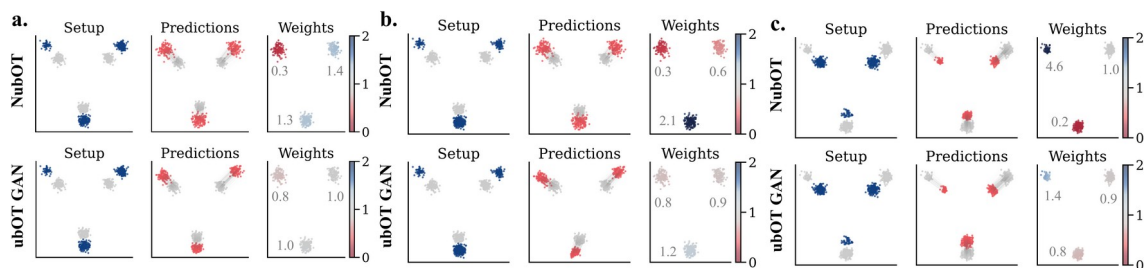


Figure 2. Unbalanced sample mapping on the synthetic benchmark. In each scenario (a, b, c) the source (gray) and target (blue) share structure but differ in shift and per-cluster sampling proportions. NubOT and ubOT GAN predict the locations (middle) and weights (right) of the transported samples; the number beside each cluster is its mean predicted weight. Both map to the right locations, but NubOT predicts the weights needed to match the target far more accurately.

Forecasting real drug responses

The real test is a single-cell melanoma drug-response dataset, generated with the imaging technology 4i. It tracks 2 co-cultured melanoma cell lines, distinguished by the mutually exclusive protein markers MelA and Sox9, as they respond to 25 different drugs. Cells are measured at 0h, 8h, and 24h; the control cells exposed only to a vehicle serve as the source, and the perturbed populations at later timepoints serve as the targets. The evaluation metric is a weighted version of kernel maximum mean discrepancy (MMD) between the predicted and observed perturbed cells, where for NubOT and the unbalanced GAN this discrepancy is weighted by each cell's predicted weight before the two populations are compared.

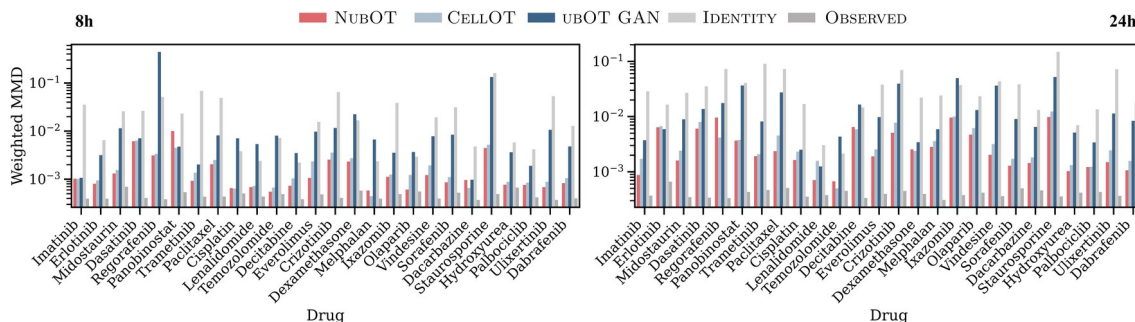


Figure 3. Distributional fit (weighted kernel MMD, lower is better) of predicted to observed perturbed cell states for every drug, at 8h (left) and 24h (right). NubOT (red) is compared against CellOT, the ubOT GAN, an Identity baseline, and Observed (a permutation of the observed cells, a rough lower bound). NubOT is at or below every baseline in almost all of the 25 perturbations.

Across the real task NubOT outperforms every baseline, including CellOT and the prior state-of-the-art unbalanced GAN, in almost all of the 25 drug perturbations. And it does so while getting the mass changes right, not merely the feature mapping, which is the combination that sets it apart from methods that nail the locations but misestimate the reweighting.

Do the predicted weights mean anything biologically?

A distributional metric can be gamed, so the more demanding question is whether the predicted mass changes correspond to real biology. The authors test this by summing NubOT's predicted weights within each subpopulation and comparing against independently observed cell counts. At 8h the agreement is very strong, with a correlation of $R = 0.95$ ($P = 2.2e-25$). At 24h it drops to $R = 0.44$ ($P = 0.004$): for some drugs the induced cell death is so severe that too few cells remain to evaluate reliably, yet the overall trend still holds.

Table 1. Predicted-weight vs observed cell-count correlation.

Timepoint	Correlation R	P-value
8h	0.95	2.2e-25
24h	0.44	0.004

Summed predicted weights per subpopulation vs the observed change in cell counts. The strong agreement at 8h weakens at 24h as severe drug-induced death shrinks the observable populations.

Figure 4 goes further and looks at where the weights land in feature space. The predicted weights track the proliferation marker Ki67 (weights above 1 where cells are dividing) and the apoptosis marker (weights below 1 where cells are dying). Under Ulixertinib, for instance, NubOT predicts proliferation in the MelA-positive population and increased cell death in the Sox9-positive population by 24h, matching the experimental observations. The model is recovering real proliferation and death signals rather than fitting the distribution superficially.

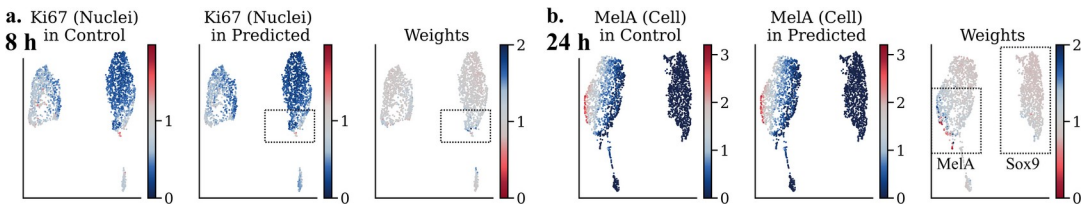


Figure 4. UMAP projections of control cells under Ulixertinib at (a) 8h and (b) 24h, colored by observed and predicted marker values and by predicted weights. NubOT predicts weights above 1 for proliferating cells in the MelA-positive population, and weights below 1 for increased death in the Sox9-positive population at 24h, consistent with the experimental cell-count measurements.

Takeaway

The lesson of NubOT is that unbalanced optimal transport can be made both practical and biologically faithful at once. By reformulating it through learnable semi-couplings and training the maps and reweighting functions with a cycle-consistent alternating scheme, it predicts, at the level of individual cells, both the movement of mass and its creation or destruction. It beats previous neural OT methods at forecasting how cancer cell lines respond to drugs, with predicted weights that line up with known proliferation and death markers.

The limits are worth stating too. The reweighting correlation weakens by 24h, exactly when drug-induced death thins the observable populations, so the most lethal regimes stay the hardest to validate, and the results here center on one melanoma system imaged by one technology. Still, explicitly modeling cell birth and death, rather than assuming mass is conserved, is a concrete step toward forecasting heterogeneous treatment responses faithfully.